



BESS **Energy Availability** Deep Dive

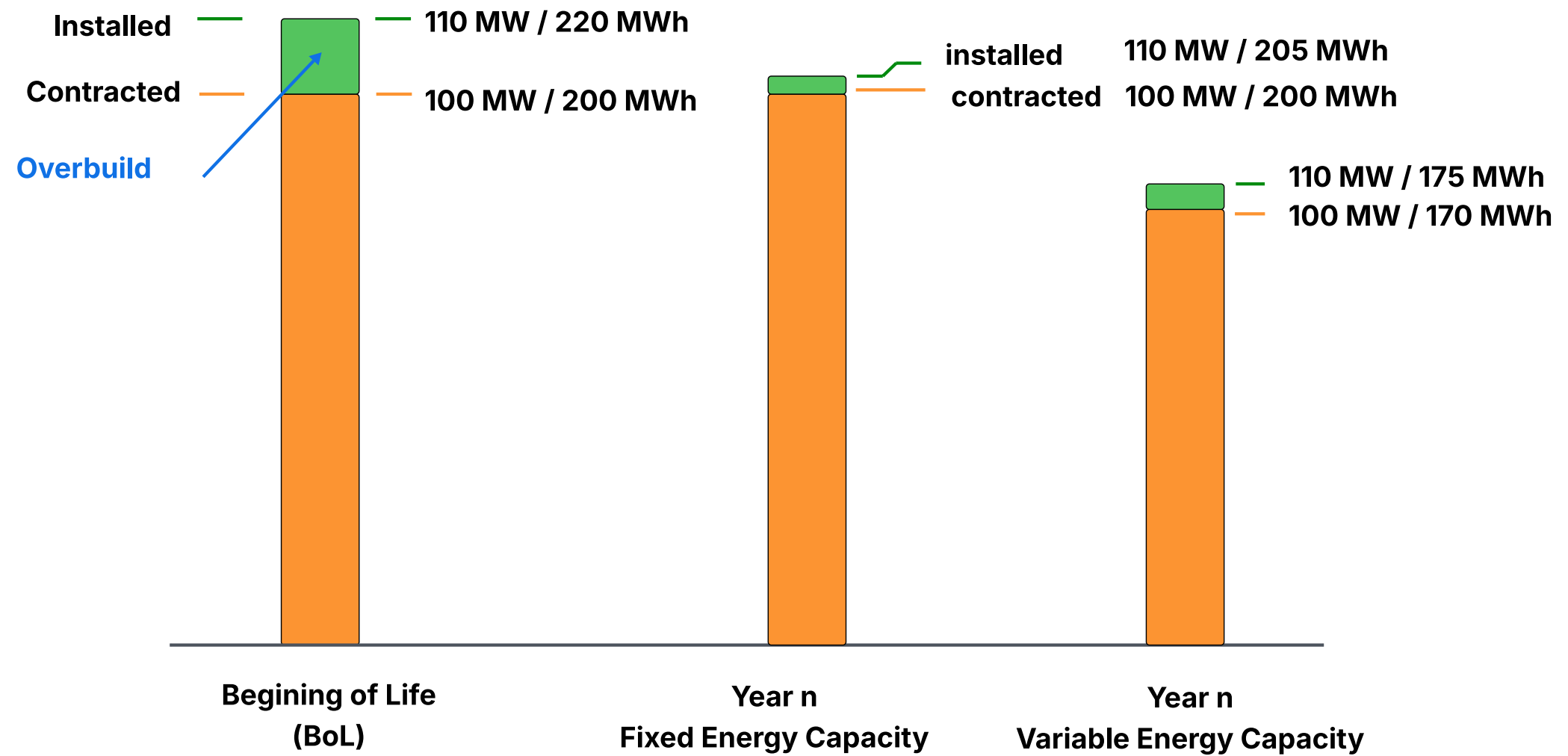
How to incorporate Energy Capacity into BESS
Availability

Module 1 - BESS Availability Explained

 by Fred Rivollier



BESS Energy Capacity Types



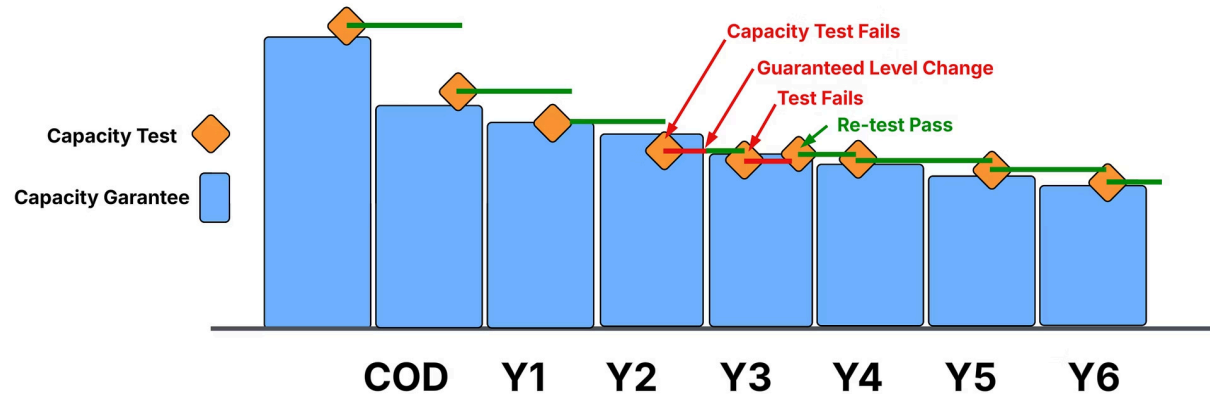
BESS Availability Types Recap

Combining all 4 availability type will form **OBE (Overall Battery Effectiveness)**

<u>Availability Type</u>	<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
① Equipment Availability	MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted
② Operational Availability	HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted
③ Power Availability	HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model.	MW	%_installed %_contracted
④ Energy Availability	HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	HIGH Measured by a controlled test	MEDIUM Capacity Tests are often best case operating conditions	MWh	%_installed %_contracted

④ Energy Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	HIGH Measured by a controlled test	MEDIUM Capacity Tests are often best case operating conditions	MWh	%_installed %_contracted



Energy Capacity Is a Slow-Moving Signal

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Energy Capacity Changes Slowly.

Energy capacity changes over **months and years** - not minutes or hours.

Offline Strings Don't Reduce Site Capacity.

A single offline string doesn't reduce site capacity - power rebalancing recovers it **within minutes**.

Equipment Offline?

Already captured in **Equipment Availability**. No need to layer it again.

Power Derate from SOC Imbalance?

Already captured in **Power Availability**.

Set Point Shortfall from Reduced Capacity?

Already captured in **Operational Availability**.

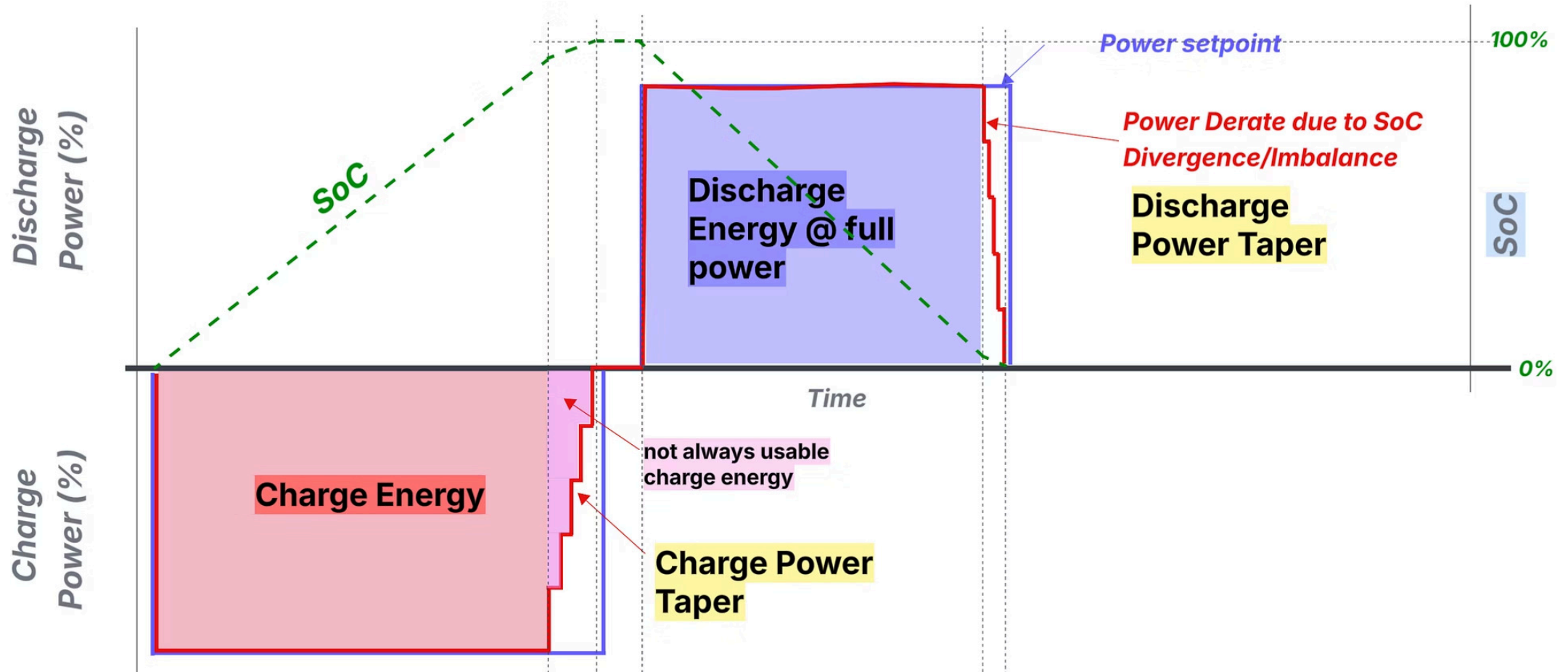
Real-Time Measurement Is Unreliable

You can't measure energy capacity in real time — only **capacity testing** gives you a reliable number.

Interval-by-Interval Layering = Double-Counting

Attempting to layer it interval-by-interval would **double-count** what the other three metrics already see.

BESS Energy Capacity Test Example



What We Actually Layer Into the Overall Availability

The only energy dimension **not already captured** by Equipment, Operational, or Power Availability is **long-term capacity retention**; the slow, permanent fade that no rebalance can fix.

We measure it through **proper capacity testing**, not telemetry.

When a site has experienced meaningful capacity loss, that retention factor gets applied as a **scalar across the OBE calculation**; a correction that reflects what the asset can truly deliver over a full cycle against the expected / guaranteed value.

It doesn't update every interval. **It updates when we perform a capacity test or the guaranteed value changes.**

Equipment Availability

Captures offline assets and hardware failures

Power Availability

Captures SOC imbalance and power derates

Operational Availability

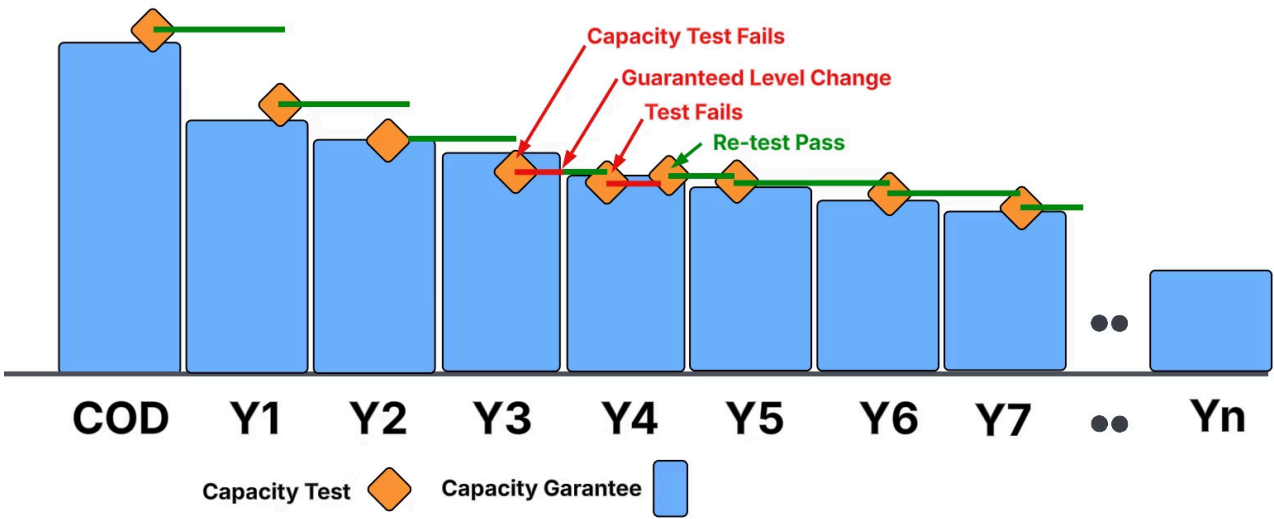
Captures set point shortfalls and dispatch gaps

Energy Availability

Captures long-term capacity retention - tested, not streamed

QA - Energy Availability %

$$QA = \min\left(\frac{Q_{\text{test}}}{Q_{\text{guaranteed}}} \times 100, 100\right) \%$$



Q_test

The energy capacity measured in the most recent controlled capacity test (MWh). Represents the actual energy the BESS can deliver under test conditions.

Q_guaranteed

The contracted or guaranteed energy capacity (MWh). The baseline against which actual performance is measured.

OBE – Overall Battery Effectiveness

Like OEE in manufacturing but built for BESS

$$OBE = \frac{1}{N} \sum_{i=1}^N \min(EA_i, OA_i, PA_i, QA_i)$$

EA

Equipment Availability

OA

Operational Availability

PA

Power Availability

QA

Energy Availability

OBE is the average, across all intervals, of the minimum availability recorded in each interval

The weakest of the four types sets the ceiling at every point in time.

What's Coming Next

In our next sessions, we'll discuss best practices to process and **automate availability calculations** and **reporting**.

Stay tuned for more Performance Engineering insights!

